Software Project Management

Project Charter

For

AuralAid: Adaptive AI Hearing with Contextual Understanding

Prepared by

Section Name: B

Student Name	Reg No
Hamza Moeen	22-SE-025
Ayesha Nazir	22-SE-079

Submitted To:

Ma'am Farwah Aizaz

DEPARTMENT OF COMPUTER SCIENCE

BS SOFTWARE ENGINEERING PROGRAM

HITEC UNIVERSITY TAXILA

Table of Contents

Contents

1. PROJECT CHARTER 1
1.1 DOCUMENT INFORMATION 1
1.2 PROJECT GOAL 1
1.3 DELIVERABLE 1
1.3.1 Research and Documentation Deliverables 1
1.3.2 Planning and Quality Assurance Deliverables 2
1.3.3 Prototype Development Deliverables 2
1.3.4 Full Development Deliverables 2
1.3.5 Testing Deliverables 2
1.3.6 Beta and Final Deployment Deliverables 2
1.4 SCOPE DEFINITION 3
1.5 PROJECT MILESTONES 3
1.6 ASSUMPTIONS 4
1.6.1 Project Scope and Requirements 4
1.6.2 Resource Availability: 4
1.6.3 Stakeholder Engagement: 4
1.6.4 Budget and Funding: 4
1.6.5 Technical Feasibility: 4
1.7 CONSTRAINTS 5
1.7.1 Time Constraints 5
1.7.2 Budgetary Constraints 5
1.7.3 Scope Control 5
1.7.4 Team Expertise 5
1.7.5 Hardware and Software Limitations 5
1.8 DEPENDENCIES 5
1.8.1 Resource Dependencies 5
1.8.2 Stakeholder Dependencies 5
1.8.3 Testing Dependencies 6
1.8.4 Regulatory and Compliance Dependencies 6
1.9 RELATED DOCUMENTS 6
1.9.1 Business Case 6
1.9.2 Statement of Work (SOW) 6
1.9.3 Project Proposal 6
1.9.4 Software Requirements Specification (SRS) 6
1.10 PROJECT ORGANIZATIONAL STRUCTURE 6
1.11 PROJECT AUTHORIZATION 7

1. Project Charter

This Project Charter formally authorizes the initiation of the project AuralAid: Adaptive AI Hearing with Contextual Understanding. It defines the project's purpose, scope, objectives, deliverables, assumptions, constraints, and responsibilities. The charter provides a foundation for planning, execution, monitoring, and control throughout the project lifecycle and serves as the primary reference document for stakeholders and the development team.

1.1 Document Information

Table 1: Document Information

Project Name:	AuralAid
Date:	05-Nov-2025
Revision Number:	V 1.0

1.2 Project Goal

The goal of this project is to design and develop AuralAid, an AI-powered adaptive hearing assistance system that enhances human auditory perception using artificial intelligence and contextual understanding. The system aims to improve speech clarity, environmental sound recognition, and adaptive filtering for users in real-world environments. The project integrates AI models, signal processing, and human-centered design to create an intelligent assistive technology solution with real-world healthcare and accessibility impact.

1.3 Deliverable

1.3.1 Research and Documentation Deliverables

- **Business Case:** Justify the project by demonstrating the need, feasibility, and expected impact.
- **Project Charter:** Define objectives, scope, and responsibilities.
- **Statement of Work (SOW):** Detail project scope, timelines, and responsibilities.
- **Software Requirements Specification (SRS):** Document all functional and non-functional requirements.

1.3.2 Planning and Quality Assurance Deliverables

- **Software Quality Assurance (SQA) Plan:** Define testing methods, quality standards, and validation procedures.
- **Software Configuration Management (SCM) Plan:** Establish version control, change management, and configuration management.

1.3.3 Prototype Development Deliverables

- **Basic AI Audio Processing:** Implement initial sound detection, filtering, and amplification.
- **Prototype Application:** Deliver a functional prototype demonstrating core AI features.

1.3.4 Full Development Deliverables

- **Feature Integration:** Add adaptive sound filtering, contextual awareness, and user controls.
- **UI/UX Implementation:** Develop a clear, accessible, and user-friendly interface.
- **Audio Feedback and Voice Integration:** Include voice guidance and interactive feedback.
- **User Personalization:** Enable basic customization of hearing preferences and modes.
- **Testing and Bug Fixing:** Ensure the system works reliably across intended use cases

1.3.5 Testing Deliverables

- Conduct unit, integration, system, and user acceptance testing.
- Provide a fully functional, tested application free from critical issues.

1.3.6 Beta and Final Deployment Deliverables

- **Beta Version Release:** Deliver for initial testing and stakeholder feedback.
- **Final Deployment:** Release final application with installation packages, documentation, and support setup.

1.4 Scope Definition

Included:

- Adaptive AI audio processing for different contexts (e.g., speech, background noise).
- Real-time sound filtering and amplification.
- User-friendly interface for configuration and monitoring.
- Voice feedback and contextual alerts.
- Personalization options for hearing preferences.

Excluded:

- Hardware development (e.g., physical hearing devices).
- Cross-platform development outside PC/mobile app scope.
- Full integration with third-party hearing devices initially.
- Multiplayer or social features.

1.5 Project Milestones

Milestone	Date	Description
Project Approval	24-Oct-2025	Obtain approval for scope, objectives, and deliverables.
SRS Submission	17-Jan-2026	Complete and review Software Requirements Specification.
Prototype Completion	17-Jan-2026	Deliver functional AI prototype.
Core Feature Finalization	21-Feb-2026	Integrate adaptive filtering, contextual awareness, and feedback.
UI/UX Completion	19-Mar-2026	Complete interface and personalization options.
Audio & Voice Integration	04-Apr-2026	Implement voice guidance and sound feedback.
Working Application Completion	10-Apr-2026	Deliver working version for testing.

Testing Completion	15-Apr-2026	Complete all testing phases.
Beta Approval	06-May-2026	Approve beta version for release.
Final Release	19-May-2026	Deliver final application and documentation.
Support System Setup	25-May-2026	Launch distribution and support setup.

1.6 Assumptions

1.6.1 Project Scope and Requirements

All functional and non-functional requirements defined in the SRS are clear, feasible, and unlikely to change significantly during the project timeline. Features like real-time speech-to-text, emotion detection, noise alert system, and health monitoring are achievable within the 10-month timeframe.

1.6.2 Resource Availability:

All required hardware (ESP32-S3, MEMS microphones, MAX30101, TMP117 sensors) and software tools (Arduino IDE/ESP-IDF, Flutter, TensorFlow Lite) will be accessible throughout development. Team members will dedicate sufficient time to design, prototype, and test the system efficiently.

1.6.3 Stakeholder Engagement:

The project advisor and potential end-users will provide timely feedback at key milestones. Beta testers will actively participate during system testing, providing actionable input on usability, accuracy, and overall performance.

1.6.4 Budget and Funding:

Sufficient funds will be available for purchasing hardware components, software licenses, and cloud services necessary for prototype development.

1.6.5 Technical Feasibility:

The selected hardware and software stack (ESP32-S3, Raspberry Pi 4, MEMS microphones, Flutter mobile app) is capable of supporting AI-driven audio processing, emotion detection, and health monitoring in real-time. Team familiarity with these tools ensures minimal technical risks.

1.7 Constraints

1.7.1 Time Constraints

The project must be completed within a 10-month timeline with milestone-based reviews for design, development, testing, and final demonstration.

1.7.2 Budgetary Constraints

Project funding is limited and must cover hardware, software, cloud services, and testing resources without exceeding the allocated amount.

1.7.3 Scope Control

Project features must remain within defined boundaries. No additional functionalities outside the SRS (e.g., cross-platform app or advanced AI features beyond the scope) will be implemented to avoid scope creep.

1.7.4 Team Expertise

Team members are new to embedded AI and wearable development, which may limit the complexity of advanced features. All modules must prioritize feasibility and integration simplicity.

1.7.5 Hardware and Software Limitations

Development and testing will be constrained by the capabilities of available microcontrollers, sensors, and mobile devices. System performance must be optimized for low-latency, real-time operation on ESP32-S3 and Raspberry Pi.

1.8 Dependencies

1.8.1 Resource Dependencies

Access to functional ESP32-S3 boards, Raspberry Pi 4, MEMS microphones, health sensors, and reliable mobile devices is critical. Continuous internet access is required for cloud synchronization, library updates, and research.

1.8.2 Stakeholder Dependencies

Timely input and approval from the project advisor, end-users, and beta testers are essential for validating design decisions and confirming that functional requirements meet user expectations.

1.8.3 Testing Dependencies

Testing phases depend on the availability of volunteers to provide real-world scenarios and feedback, as well as secure access to mobile and embedded platforms for debugging and validation.

1.8.4 Regulatory and Compliance Dependencies

The project must comply with university guidelines, privacy considerations, and intellectual property laws regarding sensor usage, AI models, and software libraries.

1.9 Related Documents

1.9.1 Business Case

Provides justification for developing an AI-enabled, low-cost hearing aid system and highlights potential social and commercial benefits.

1.9.2 Statement of Work (SOW)

Details the project deliverables, milestones, responsibilities, and timelines for successful completion of the AuralAid system.

1.9.3 Project Proposal

efines objectives, scope, and expected outcomes of the Smart Hearing Aid System, serving as a foundation for detailed project planning and execution.

1.9.4 Software Requirements Specification (SRS)

Outlines all functional (speech-to-text, emotion detection, noise alerts) and non-functional (latency, accuracy, reliability) requirements for the system.

1.10 Project Organizational Structure

Table 2: Organizational Structure

Function	Name	Role
Project Manager	Hamza Moeen	Oversees the project, coordinates tasks, manages timelines, and ensures milestones are met.

Embedded Systems & AI Developer	Hamza Moeen	Implements ESP32-S3 firmware, DSP algorithms, and AI models for speech enhancement and emotion detection.
Mobile App Developer	Ayesha Nazir	Designs and develops Flutter-based mobile interface, integrating live captions, emotion visualization, and alerts.
Data/AI Specialist	Ayesha Nazir	Optimizes speech-to-text and emotion detection models for low-latency processing.
Project Advisor	Mr. Junaid Ali Khan	Provides technical guidance, monitors progress, and ensures adherence to academic standards.
End-User / Beta Tester	Volunteers	Provide usability feedback, performance evaluation, and system validation in real-world scenarios

1.11 Project Authorization

Approved by	Name	Date
Business Manager	Hamza Moeen	06-Nov-2026
Project Manager	Ayesha Nazir	06-Nov-2026

This formal approval confirms authorization to proceed with the development of the AuralAid Smart Hearing Aid System.